Sanity Checks for Reinforcement Learning Explanations

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Abstract. Interpretability is proposed as a basis for oversight: if we can see which inputs drive a policy, we can decide whether to trust it. That argument needs attributions to carry information about the agent, and we show they need not. On an empirically efficient prediction market, a PPO agent whose return is statistically indistinguishable from abstention produces attributions with cross-seed rank correlation 0.849 and unanimous agreement on the most important feature; every conventional check passes and the explanation is empty. We give a null-model test built on the additivity identity attribution schemes already satisfy, comparing an agent's attribution span against agents trained where the observation channel carries no information. It rejects a planted signal at $z = 12.27$, fails to reject the market agent at $z = +0.23$, and fires at $z = +3.40$ on a genuinely learnable task built from the same real episodes. Explanations are also cheap to forge: an auxiliary training penalty drives a feature's attribution from 39.9% to 3.2% of total mass at no measurable cost in return, with no deception and no attack on the explainer. The established sanity check for this purpose, randomising network parameters, does not transfer: across sixteen checkpoints on four control tasks it never exceeds $z = 2.45$, because its reference distribution equals the spread of random-initialisation return to three significant figures and so measures the variance of initialisation, not the explanation.

1 Introduction

Interpretability is proposed as a basis for oversight: if we can see which inputs drive a policy, we can decide whether to trust it. This requires that an attribution mean something, and attribution methods return an answer regardless. Given any policy and state, Shapley values [6, 9] and integrated gradients [11] produce a ranked, signed vector. Nothing in the output separates an attribution reflecting structure the agent exploits from one reflecting nothing.

Supervised learning handles this with randomisation tests. Adebayo et al. [1] compare explanations of a trained model against a model with randomised parameters, and against a model trained on permuted labels. Reinforcement learning has adopted only the first. Huber et al. [5] apply the parameter test to Atari saliency and observe that the data-randomisation test has no obvious RL analogue: there is no labelled dataset to permute.

We supply that analogue, and show the half already adopted does not work.

Contributions.

1. An environment-level null for RL: corrupt the observation channel, leaving dynamics and reward intact, so the agent trains normally with nothing to condition on. No access to task internals is required.
2. Evidence that parameter randomisation has near-zero power in RL.
3. A ground-truth validation protocol using environments where the correct attribution is known by construction.
4. A demonstration that attribution is not identified by behaviour.

What we do not claim. The argument that RL attribution should target outcomes rather than policy outputs is due to Beechey et al. [3]; FastSVERL [2] addresses the scalability, off-policy and temporal problems we only name. Our test adapts Adebayo et al. [1]. Deletion-based fidelity is established practice in XRL [4, 7]. Our contribution is the RL construction, the demonstration that the incumbent fails, and the consequences.

2 Method

Let $v(S)$ be the expected episode return of an agent observing only the features in $S \subseteq N$, with features outside S replaced at every timestep by draws from a reference distribution. Define the *attribution span* $\Delta = v(N) - v(\emptyset)$.

Shapley values and integrated gradients satisfy additivity identities under which per-feature attributions sum to Δ, so the span can be read off either. It is worth stating that Δ is *not* a Shapley quantity: it is the difference between two masked rollouts and depends only on the masking. This has a practical consequence: it costs two evaluations rather than $2^{|N|}$.

Blind-observation null. *Replace every observation with an independent draw from a fixed distribution matched to the environment's observation moments. Leave dynamics and reward untouched. Train normally* (Figure 1).

Such an agent faces the real task and receives real return but cannot condition on anything. Unlike parameter randomisation it yields a *normally trained* policy: the failure mode that occurs in deployment. Matching the reference moments matters, or the network fails for reasons of scale rather than information.

Given null draws with mean $\bar{\Delta}$ and standard deviation

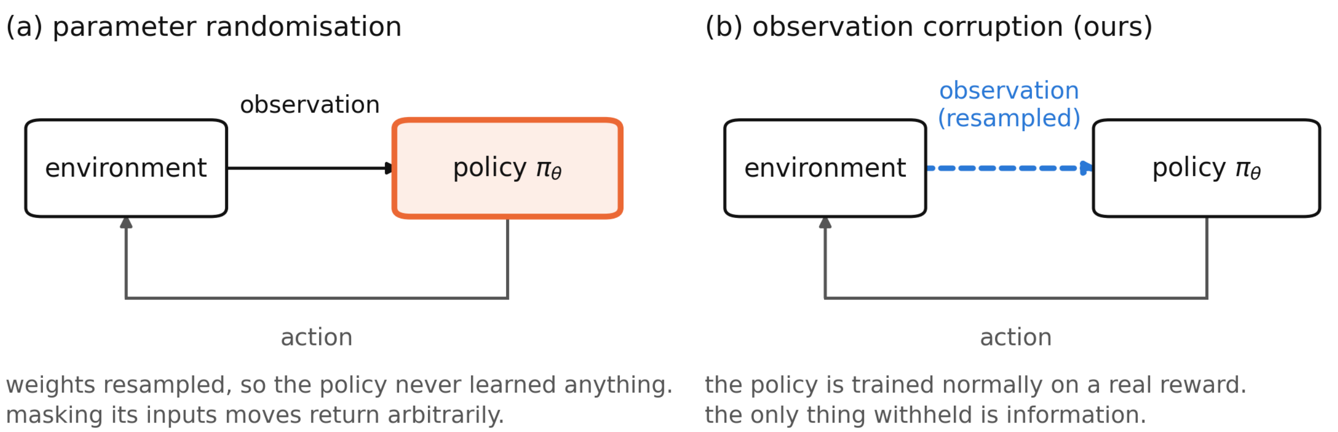

Figure 1: Two ways to build a reference distribution of explanations for an agent that learned nothing. The established check (a) randomises the policy's parameters, leaving a network that was never trained. Ours (b) corrupts the environment's observation channel instead, so the agent optimises a real reward with nothing to condition on. Only (b) yields a normally trained policy, which is the failure mode that occurs in deployment.

s we report a rank p-value and $z = (\Delta_{\text{obs}} - \bar{\Delta})/s$, and require both to reject. The rank statistic assumes no shape but bottoms out at $1/(n+1)$; the normal statistic has resolution but assumes a shape the null need not have. If every null draw is identical, $s = 0$; we return $z = \pm\infty$ and surface the degeneracy rather than guarding it away, but treat it as a reason to distrust the reference. A point-mass null has no specificity, and building one is how a more principled-looking null construction turned out to fire on a corpus with no signal in it (Section 4.6).

3 Setup

Environments with known answers. Explanation evaluation lacks ground truth [4]. We construct fixed-length synthetic corpora where the answer is fixed by construction: a *planted signal* (one of 18 features informative about the outcome), a *null* (none), and a *decoy* (one feature correlating 0.989 with the outcome in-sample and 0.006 out of sample).

Market. Our main setting is a binary contract settling to \$1.00 if BTC is higher after 15 minutes, on a regulated prediction market: 6,425 settled contracts over 68 days, walk-forward split by time, test evaluated once. Every episode resolves to a known value. Agents are trained with PPO [8]; reward is the change in mark-to-market equity net of the exchange's published fee schedule, a 9% round-trip hurdle at the price this contract trades at.

Control tasks. CartPole-v1 and Acrobot-v1, whose observation spaces are small enough to enumerate all coalitions, so Shapley values there are exact.

4 Results

4.1 An explanation that passes every check

The market is empirically efficient: its implied probability has a weighted mean absolute calibration error of 0.0172, and spot return correlates 0.476 with the outcome but only 0.016 with the residual after the price is accounted for. No policy beats abstention (Table 1).

Table 1: Test split, 1,284 held-out episodes. Paired bootstrap against always-flat.

Policy	Mean P&L	Trades/ep	p vs flat
always-flat	+0.000	0.00	n/a
buy-and-hold	−1.608	1.00	0.23
PPO (5 seeds)	$\mathbf{-0.595 \pm 0.176}$	1.37	**0.10**
mean-reversion	−11.615	4.94	<0.0001
random	−18.532	9.33	<0.0001

Table 2: Null-model test. Null from 24 blind-trained agents, span $+8.19 \pm 5.11$.

Case	Span	z	Verdict
planted signal	+70.89	**+12.27**	informative
null corpus	+11.86	+0.72	not distinguishable
real market	**+9.37**	**+0.23**	**not distinguishable**

PPO's shortfall is its own: on synthetic data with no signal by construction the same agent scores -0.47 ± 0.47. Yet across five seeds whose pairwise performance differences are indistinguishable in all ten pairs, its attributions have cross-seed rank correlation 0.849 with 100% agreement on the top feature, a top-5 ranking whose adjacent gaps exceed their combined standard errors, and a legible story: time-to-expiry dominates, reading as an agent that learned when not to trade.

The null test disagrees (Table 2, Figure 2). It has power and specificity, and the market agent sits at the centre of the distribution of explanations of nothing.

We note one consequence for our own reporting. An earlier draft stated that "the whole observation is worth +5.914 per episode against being blind" and presented it as a finding. Blind agents produce $+8.19 \pm 5.11$. It was noise.

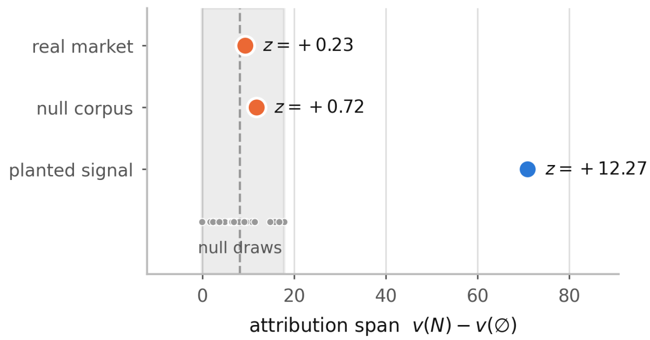

Figure 2: Attribution spans against the null. The band is the range over 24 agents trained where the observation channel carries no information.

Table 3: Null construction on four control tasks. MountainCar is degenerate under both: no policy in the budget reaches the goal.

Environment	Parameter	Environment	Ratio
CartPole	**±138.92**	**±3.29**	42×
Acrobot	**±124.94**	**±0.00**	degenerate
MountainCar	**±0.00**	**±0.00**	0×
Pendulum	**±131.42**	**±37.79**	3×

Table 4: Steering one feature's attribution.

Corpus	Attribution	Return	p
market	39.9% → **3.2%**	−3.49 → −1.65	**0.66**
planted signal	46.5% → **1.5%**	+45.04 → **+7.92**	<0.001

4.2 A positive control on real data

Every case above where the test fires is synthetic, which invites the objection that it detects planted signal rather than information. Holding the corpus, features, normalizer, split and null construction fixed and varying only the objective answers it. The market's price is calibrated to 0.0172, so it is highly informative *about the outcome* while offering nothing to trade against once a 9% round-trip fee is paid. Scoring the same architecture on the same real episodes for *calling* the settlement rather than trading it, against a null of 24 agents trained on those episodes with observations blinded, gives a span of +7.46 against -0.48 ± 2.34: $z = +3.40$, informative, unchanged in all 4,000 resamples. The trading agent on identical episodes and an identical null gives +7.41 against $+2.47 \pm 2.29$, $z = +2.16$, which the two-part rule declines, though only 64% of resamples agree and we report it as borderline. The two spans are nearly the same size; what separates them is the null, since a blind agent cannot call the outcome at all but can still decline to trade. The market data is not uninformative. The information is not monetisable, and an attribution of the trading agent cannot tell those two situations apart.

4.3 Validation against ground truth

On the planted corpus, outcome attribution ranks the planted feature first with 55% of total mass. On the decoy corpus, an agent trained where the decoy predicts the outcome (in-sample +47.24, held out −6.83) is explained on episodes where the decoy is provably worthless: per-decision attribution gives it 6.5% of mass at rank 3 of 18, outcome-based attribution 1.1% at rank 10. Both are true; the policy does key on the decoy. Per-decision attribution is correct about behaviour and misleading about value.

4.4 The established null does not transfer

We build both nulls for the same task (Table 3, Figure 3). A randomly initialised policy behaves arbitrarily, so masking its inputs moves return unpredictably and the reference distribution is very wide. Across sixteen checkpoints the parameter null never exceeds $z = 2.45$, including on a converged CartPole policy scoring 404 of 500, while the environment null reaches $z = +117.6$. The two disagree on five of the sixteen.

Why it is wide. The gap has a mechanism, and it is not about explanations. On every task the parameter null's width equals the spread of random-initialisation return to three significant figures: 138.38 against 138.92 on CartPole, 123.04 against 124.94 on Acrobot. In $\Delta = v(N) - v(\emptyset)$ the masked term is near-constant for a random network (sd 1.8% of the unmasked term's on CartPole), because a policy not using its observations loses little when they go. The span inherits the unmasked variance entire: *parameter randomisation measures the variance of initialisation*, which is why a test built on it has no power.

Acrobot also shows the test tracking *whether there is anything to explain* rather than agent quality: its agent fails completely until it learns, the span is exactly 0.00 there, and the test correctly declines. CartPole's observations matter at every skill level and the test fires throughout. We do not claim undertrained agents generally produce empty explanations.

4.5 Attribution is not identified by behaviour

We add a training penalty on the divergence between $\pi(\cdot \mid s)$ and $\pi(\cdot \mid s')$, where s' has one feature resampled from the batch marginal: the same perturbation the attribution measures (Table 4).

The attribution is equally suppressible in both cases. What differs is the price: suppressing the planted signal destroys 81% of return; on the market it costs nothing measurable. Explanations are steerable at no cost exactly where they are not tracking anything, corroborating Section 4.1 by a route sharing no machinery with the null test. The control matters: had steering succeeded on both, the penalty would merely be defeating the attribution method.

This differs from Slack et al. [10], who construct classifiers that detect the off-manifold points LIME and SHAP query. Here the agent is trained normally and the result-

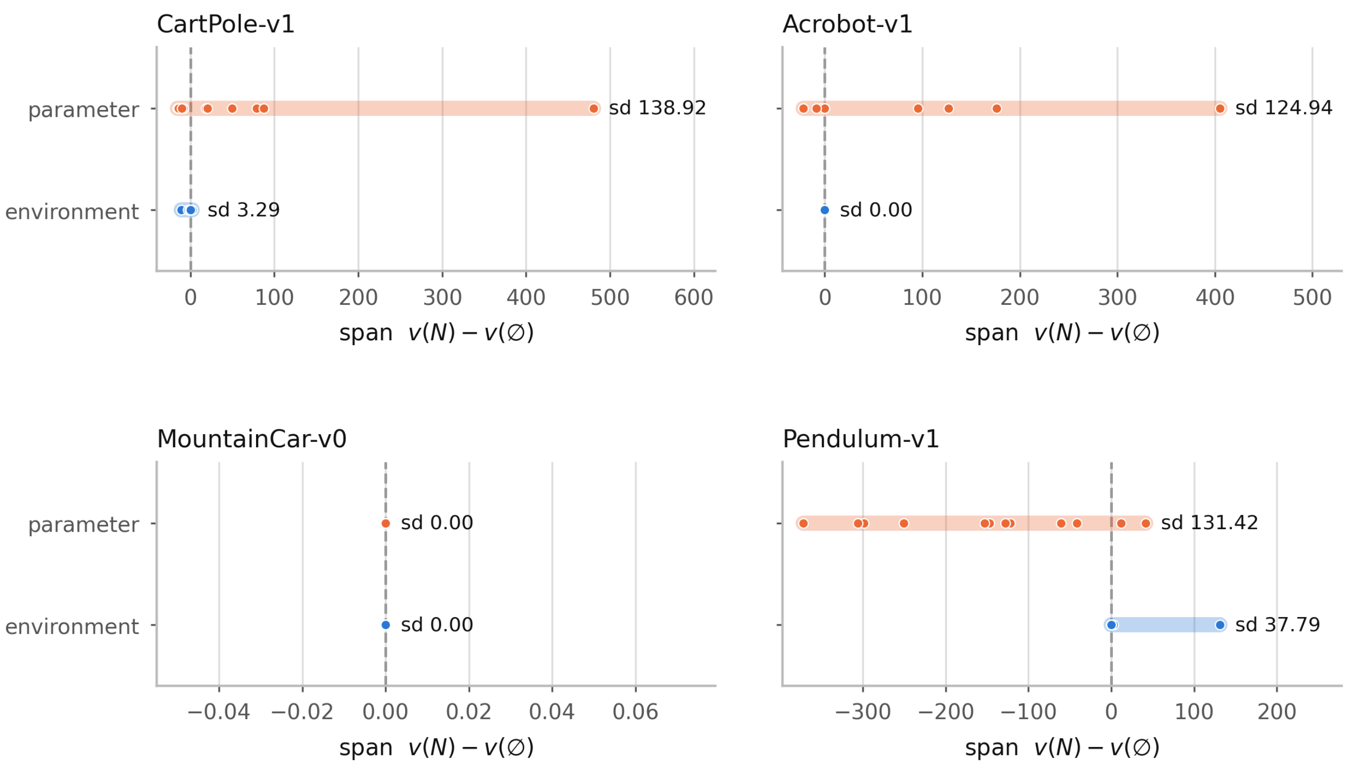

Figure 3: Spread of each null construction. Width determines power: a very wide reference distribution cannot reject anything.

ing explanation is *correct* about the resulting model; the point is that behaviour does not pin it down.

4.6 What the corpus-matched null costs

Our null trains its agents on synthetic signal-free corpora, so the corpus varies along with the information. Holding the corpus fixed instead, by blinding the observation channel of the real episodes, looks strictly better and is what the definition above literally says. Run on every case it produces a false positive: on a corpus built to contain no informative feature, span $+13.90$ against a null of 0.00 ± 0.00, which fires. Its verdict on the real market also flips with the null agents' training budget, from $z = +0.55$ at 20 updates to $z = +4.23$ at 80, as those agents converge.

The mechanism is that the two nulls are not the same null. A signal-free corpus gives agents that can see but have nothing to see; they still respond to observational structure and their spans are spread out. Blinding gives agents that cannot see; they converge to a constant policy, whose return does not depend on its inputs at all, so every null span is exactly zero. The agent under test is sighted, so the right reference is sighted agents with nothing to learn. We keep the signal-free-corpus null, whose fault is that it widens the reference and therefore errs toward declining.

5 Limitations

The test adapts Adebayo et al. [1]; the outcomes framing is Beechey et al. [3]'s. The span measures information in aggregate, not whether the decomposition among features is right. Temporal credit assignment and off-policy attribution are named, not solved; these are FastSVERL's [2] territory. An earlier mechanism we proposed for the 42× gap (that randomising weights perturbs the measurement's domain and compounds along the trajectory) was tested and rejected: over horizons 4 to 28 both nulls grow at indistinguishable rates ($\alpha = 0.94$ against 0.83). The initialisation-variance identity accounts for it instead, and involves no compounding. Four environments is too few to fit a law, so we report that correspondence rather than a coefficient. Scope is one real domain and four classic-control tasks, all with small observation spaces, so nothing here speaks to pixel-based agents.

6 Conclusion

An attribution can be stable, precisely estimated, plausible, and empty. The checks practitioners apply do not detect this, because none of them ask whether the explained agent learned anything. A null-model test does, using a statistic the attribution framework already provides. The established test for this purpose does not survive the move to reinforcement learning.

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